

3. Project Design Phase

Introduction

The Project Design Phase focuses on designing the overall structure of the AI-Powered Smart Visitor Management System. In this phase, the system architecture, database structure, application workflow, and dashboard design are defined. The design ensures that the system is scalable, secure, and easy to use.

The application is developed using the ServiceNow platform and uses built-in modules such as App Engine Studio, Flow Designer, Business Rules, Client Scripts, Reports, and Dashboards.

System Architecture

The Smart Visitor Management System follows a structured workflow where visitor information is collected, validated, processed, and displayed through reports and dashboards.

Architecture Flow

Visitor → Registration Form → Visitor Request Table → Business Rules → Flow Designer → Email Notification → Reports → Dashboard

Architecture Description

Visitor Registration Form

The visitor enters details such as Visitor Name, Mobile Number, Email, Purpose of Visit, and Visit Date through the registration form.

Validation Layer

Client Scripts validate the entered information before submission. Mobile numbers are checked for proper format, and visit dates are verified to prevent past dates from being selected.

Visitor Request Table

After validation, the data is stored in the Visitor Request table. This table acts as the central repository for all visitor-related information.

Business Rules Layer

Business Rules automatically execute when records are created. These rules set default values and generate AI recommendations based on the purpose of the visit.

Flow Designer Layer

Flow Designer automates business processes such as sending email notifications and updating request status.

Reporting Layer

Reports are generated from visitor records and provide insights into visitor activities.

Dashboard Layer

The dashboard displays graphical reports and analytics for management review.

Database Design

The database design defines how visitor information is stored and managed within ServiceNow.

Table Name

Visitor Request

This custom table stores all visitor information and system-generated details.

Field Definitions

Visitor ID

Type: Auto Number

Purpose: A unique identifier generated automatically for every visitor request.

Example:

VIS0001023

Visitor Name

Type: String

Purpose: Stores the full name of the visitor.

Mobile Number

Type: String

Purpose: Stores the visitor's contact number.

Validation: Only 10-digit mobile numbers are accepted.

Email

Type: Email

Purpose: Stores the visitor's email address for notifications.

Purpose

Type: Choice

Purpose: Stores the reason for the visit.

Available Choices:

- Interview
 - Client Meeting
 - Business Meeting
 - Delivery
 - Training
 - Vendor Visit
 - Other
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Visit Date

Type: Date

Purpose: Stores the scheduled date of the visit.

Status

Type: Choice

Purpose: Tracks the current status of the visitor.

Available Status Values:

- Requested
- Approved
- Rejected
- Checked In
- Checked Out

- Cancelled

Table - Visitor Request

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Application

* Name Remote Table

Columns Controls Application Access

Table Columns for text Search

Column label	Type	Reference	Max length	Default value	Display
AI Recommendation	String	(empty)	500	false	
Status	Choice	(empty)	40	false	
Mobile Number	String	(empty)	40	false	
Email	String	(empty)	40	false	
Visit Date	String	(empty)	40	false	
Updated by	String	(empty)	40	false	
Updates	Integer	(empty)	40	false	
Updated	Date/Time	(empty)	40	false	
Created by	String	(empty)	40	false	
Sys ID	Sys ID (GUID)	(empty)	32	false	
Created	Date/Time	(empty)	40	false	
Purpose	Choice	(empty)	40	false	
Visitor Name	String	(empty)	100	false	
Number	String	(empty)	40	javascript:global.getNextObj(NumberOfNumberPadded);	false
Insert a new row...					

Delete Update Delete All Records

AI Recommendation

Type: String

Purpose: Stores recommendations generated automatically by the AI Recommendation Engine.

Example:

Schedule HR Meeting Room

The screenshot shows the ServiceNow user interface. At the top, a dark blue header bar contains the ServiceNow logo, navigation tabs (All, Favorites, History, Workspaces), and a notification badge for 'Email - Visitor Request Submitted'. Below the header, the main content area displays an email preview for a 'Visitor Request Submitted' event. The email details are as follows:

- Mailbox:** Outbox
- State:** Ready
- Target:** (empty)
- Subject:** Visitor Request Submitted
- Recipients:** ram@gn
- Body:**

Preview Email

Hello,

Your Visitor Request has been submitted successfully:

Visitor Name: Ram

Purpose: Interview

Visit Date: 23/06/2026

AI Recommendation: Schedule HR Meeting Room

Status: Requested

Thank you,

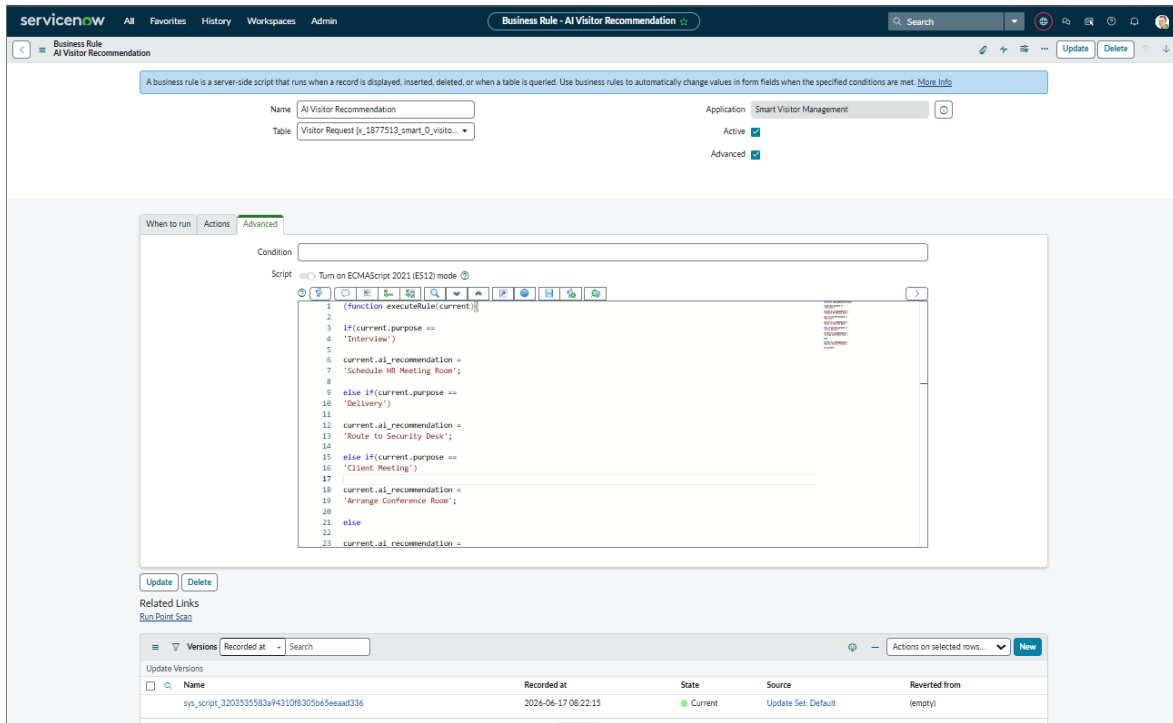
Visitor Management Team
- Content type:** (empty)
- Headers:**

Message-ID: <1724970855.5.1781713622410@app131050.ord191.service-now.com>

X-ServiceNow-Source: FlowDesigner-52226715F2d4310058c2978145ca1cb

X-ServiceNow-Generated: true

X-ServiceNow-SysEmail-Version: 2



Flow Diagram

The system workflow follows a sequence of automated processes.

Process Flow

Visitor Request Created ↓ Client Validation ↓ Store Data in Table ↓ Generate AI Recommendation ↓ Set Default Status ↓ Trigger Flow Designer ↓ Send Email Notification ↓ Check In Process ↓ Check Out Process ↓ Generate Reports ↓ Update Dashboard

Application Structure

The application is divided into several modules.

Visitor Registration Module

Used to capture visitor information through forms.

Validation Module

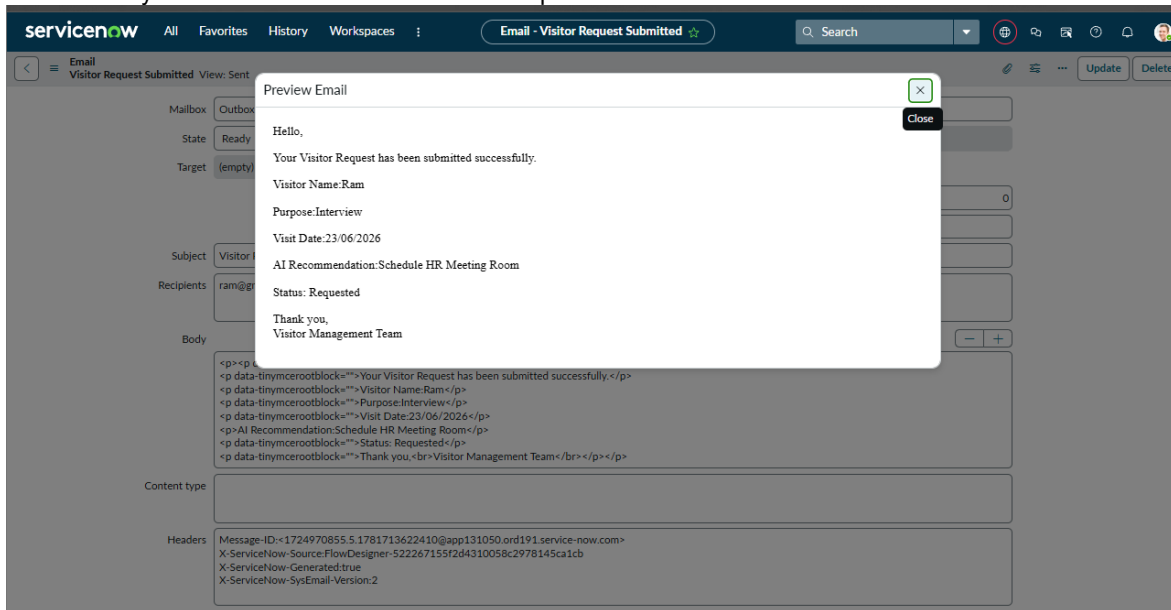
Validates mobile numbers and visit dates.

AI Recommendation Module

Generates recommendations based on visitor purpose.

Email Notification Module

Automatically sends email notifications after request submission.



Visitor Tracking Module

Tracks visitor check-in and check-out activities.

Reporting Module

Creates analytical reports from visitor data.

Dashboard Module

Displays reports and charts in a single interface.

Dashboard Design

The Smart Visitor Dashboard provides a graphical representation of visitor activities.

Visitor Status Report

Type: Pie Chart

Purpose: Displays the number of visitors in each status category such as Requested, Approved, Checked In, and Checked Out.

Visitor Purpose Report

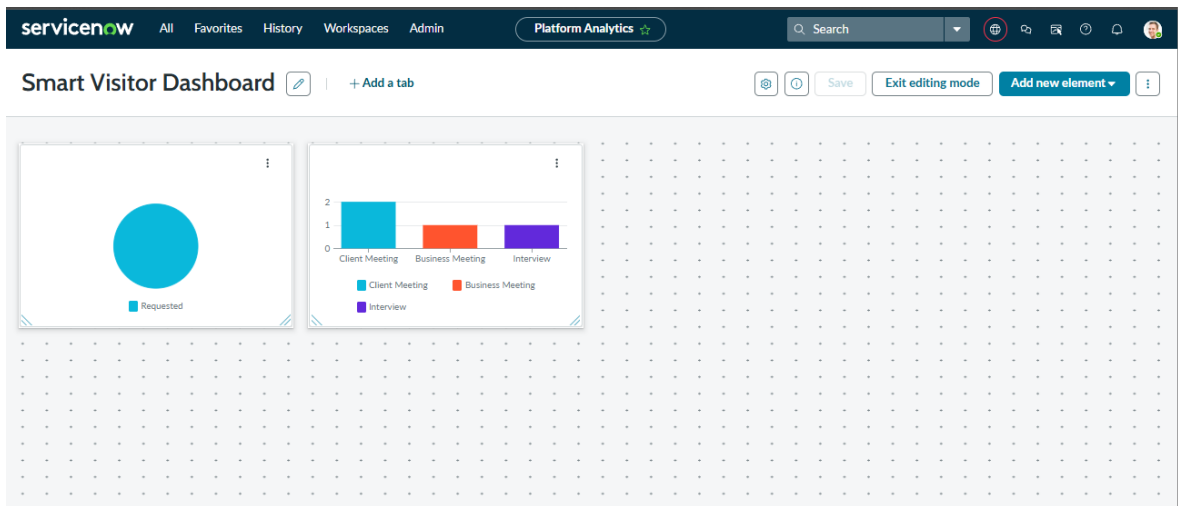
Type: Bar Chart

Purpose: Shows the distribution of visitors based on visit purpose.

Monthly Visitor Report

Type: Column Chart

Purpose: Displays monthly visitor trends and visitor growth over time.



Benefits of the Design

- Simple and user-friendly architecture.
- Automated business process management.
- Real-time visitor tracking.
- Secure storage of visitor information.
- Easy report generation.

- AI-based recommendations.
 - Scalable and maintainable system.
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Conclusion

The Project Design Phase provides a clear blueprint for the Smart Visitor Management System. The architecture integrates ServiceNow forms, tables, Business Rules, Flow Designer, Reports, and Dashboards into a single automated solution. This design ensures efficient visitor management, improved security, and better decision-making through analytics and reporting.